

Hao Lee

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SUMMARY

Master student at University of Washington, specializing in robot learning and control. With hands-on experience building VLA, reward learning, planning and optimal control for robotic manipulation.

EDUCATION

University of Washington

Master of Science in Mechanical Engineering (data science track)

Related coursework: CSE546 Machine learning / CSE571 AI-robotics / CSE579 Reinforcement Learning / EE546 control sys theory

Sep 25 – Present

Seattle, U.S.A

National Cheng Kung University

International bachelor's degree of Energy Engineering, Aerospace and Aeronautical Engineering

Sep 21 – Jun 24

Tainan, Taiwan

PROFESSIONAL EXPERIENCE

Research Assistant

Robot Learning Lab, Paul G. Allen School, UW

- Designed LLM-based reward modeling into MPAIL framework

May 26 – Present

Seattle, U.S.A

Data Engineer Intern

AUO-Digitech

- Processed and curated large-scale manufacturing datasets to build training pipelines for predictive ML models; performed hyperparameter tuning and performance evaluation using PyTorch-based frameworks. Developed NLP applications (chatbots, text classification) leveraging fine-tuned BERT and GPT models
- Supported cross-functional R&D on drone applications in factory automation, bridging ML knowledge with robotics system integration

Jun 24 – Aug 24

Hsinchu, Taiwan

RESEARCH EXPERIENCE

LLM-Inferred Reward Functions for Task-based RL

Capstone Project, UW (CSE579 Reinforcement Learning)

- Built an RLHF-style reward learning pipeline that replaces human annotators with an LLM: given environment source code, an LLM generates both a fixed reward function r_{fixed} and a semantic trajectory summarizer via RAG
- Trained an ensemble reward model ($3 \times$ MLP, Bradley-Terry loss) with active learning via disagreement scoring to minimize LLM queries; combined reward $r_{\text{total}} = r_{\text{fixed}} + g \cdot R_{\phi}$ used to train PPO on HalfCheetah-v5

Apr 26 – Present

Seattle, U.S.A

Vision-Language-Action (VLA) Robot Control Pipeline

Capstone Project, UW (CSE571 AI-Robotics)

- Built an end-to-end 4-stage VLA pipeline: natural-language command $\rightarrow$ semantic waypoint planning $\rightarrow$ 3D projection $\rightarrow$ Cartesian trajectory execution on a 6-DOF robot arm
- Integrated Google Gemini as vision-language planner to infer semantic 2D waypoints from RGB frames, then lifted pixel-space waypoints into 3D robot-frame coordinates via stereo depth projection with spatial-temporal median filtering

Jan 26 – Mar 26

Seattle, U.S.A

Multi-Drone Autonomous Minefield Mapping & Path Planning

Washington Aerial Robotics Team (LARC Mission 10))

- Architected peer-to-peer coordination layer using ROS2/FastDDS with gossip-protocol-style diffusion across 4 autonomous agents, achieving sub-100ms propagation latency
- Designed decentralized task-auction mechanism enabling agents to independently estimate costs and compete via claim broadcasts and implemented hierarchical state machine with priority-ordered behavior tree for reactive collision avoidance

Jan 26 – Present

Seattle, U.S.A

GNN Formation Control for Multi-Drone Systems

Capstone Project, UW (EE546 Advanced Control Theory)

- Designed a GNN policy trained via imitation learning to maintain chain formation across leader-follower drone networks
- Achieved zero-shot generalization from $n=4$ to $n=8$ followers using permutation-equivariant shared weights

Jan 26 – Mar 26

Seattle, U.S.A

Deep Reinforcement Learning for Quadrotor Flight Control — Undergraduate Thesis

NCKU MARS Lab (Dr. Chen)

- Implemented DDPG algorithm in simulation to train an autonomous quadcopter flight controller, targeting higher stability and faster convergence than classical PID controllers
- Performed ablation studies on reward shaping, action-space normalization, and noise injection to optimize flight policy; presented findings as pioneering exploration of DRL-based UAV control within the lab group

Sep 23 – Jun 24

Tainan, Taiwan

TECHNICAL SKILLS

Languages, Frameworks & Tools: Python, C++, MATLAB, ROS2, PyTorch, CasADi, MuJoCo, CUDA, Docker, Git, LATEX

Robotics & Control: Robot Manipulation, MPC, Path Planning, SLAM, Behavior Trees, FK/IK, Multi-Agent Coordination

ML / AI: Deep RL, GNN, Transformer, World Models, Imitation Learning, Multimodal LLM, VLA